SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**

No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:

*I **K Sathwik Kumar (192425051)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K Sathwik Kumar

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	

10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded **ANSWER ANY ONE QUESTION**

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.

5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Decision Tree Learning: A Comparative Review of ID3, C4.5 and CART

Abstract

Decision tree learning is one of the most widely used supervised learning approaches because it combines relatively simple model construction with interpretable decision rules. This review examines three foundational algorithms—**ID3, C4.5, and CART**—with emphasis on their attribute-selection methods, handling of numerical and missing data, pruning strategies, interpretability, computational characteristics, and classification performance. ID3 introduced entropy and Information Gain for recursive tree construction, while C4.5 extended the approach through Gain Ratio, support for continuous attributes and missing values, and error-based pruning. CART introduced binary recursive partitioning and the Gini-based splitting criterion together with cost-complexity pruning. (DOI) Recent research continues to show that splitting criteria, pruning, noisy labels, fairness, and computational efficiency remain important research issues. A 2023 multi-dataset study found that ID3 and CART performed strongly across several evaluation dimensions, while newer studies investigate optimized C4.5, fairness-aware trees, and more computationally sophisticated pruning methods. (DOI) The review identifies a research gap in simultaneously optimizing **accuracy, robustness, fairness, pruning efficiency, and interpretability**. A promising direction is an adaptive decision-tree framework that selects splitting and pruning strategies according to dataset characteristics while incorporating fairness and robustness constraints.

Keywords: Decision Trees, ID3, C4.5, CART, Pruning, Classification

1. Introduction

Decision trees are supervised machine-learning models that recursively divide a dataset into smaller subsets according to selected attributes. The final structure consists of decision nodes, branches, and leaf nodes representing predictions. Their major advantage is interpretability: a prediction can be followed through a sequence of understandable rules rather than through a large number of hidden mathematical parameters.

Decision-tree research has evolved through several important algorithms. **ID3**, introduced by J. Ross Quinlan, established an information-theoretic approach based on entropy and Information Gain. (DOI) **C4.5**, developed as an extension of ID3, introduced important capabilities including Gain Ratio, treatment of continuous attributes, missing-value handling, and pruning. (ScienceDirect) **CART**, developed by Breiman, Friedman, Olshen and Stone, established a binary-tree framework for classification and regression. (WorldCat)

Decision trees remain relevant because they offer a useful balance between predictive modeling and explainability. A 2024 IEEE Access survey describes decision trees as popular because of their simplicity and interpretability and reviews ID3, C4.5, CART and later tree-based methods across modern applications. (Directory of Open Access Journals)

2. Research Problem and Objectives

2.1 Research Problem

Although ID3, C4.5, and CART are all decision-tree algorithms, they use different attribute-selection and pruning mechanisms. These differences affect:

- Classification accuracy
- Tree size
- Overfitting
- Computational cost
- Handling of numerical data
- Handling of missing values
- Interpretability

- Generalization
- Robustness to noisy data

Therefore, simply stating that one algorithm is "better" than another is insufficient. Their effectiveness depends on dataset characteristics and evaluation criteria.

2.2 Objectives

The objectives of this review are:

1. To study the principles of ID3, C4.5, and CART.
2. To compare their attribute-selection methods.
3. To examine their pruning strategies.
4. To compare their interpretability.
5. To analyze computational characteristics.
6. To compare classification performance using published experimental evidence.
7. To identify limitations in existing decision-tree research.
8. To identify a research gap.
9. To propose an improved research direction.
10. To determine when each algorithm is most suitable.

3. Literature Review

3.1 ID3

ID3 was presented by Quinlan as an inductive decision-tree method. The algorithm selects attributes using **Information Gain**, which is based on entropy. At each node, the attribute providing the highest Information Gain is selected as the splitting attribute. (DOI)

Main characteristics

- Uses **Entropy**
- Uses **Information Gain**
- Originally designed mainly for categorical attributes
- Uses recursive top-down tree construction

A major weakness of Information Gain is its tendency to favor attributes with many distinct values. Recent comparative research continues to identify this attribute-selection issue as an important limitation. (DOI)

3.2 C4.5

C4.5 is an extension of ID3 developed by Quinlan. It uses **Gain Ratio** rather than raw Information Gain. Gain Ratio was designed to reduce the bias toward attributes containing many possible values. (DOI)

C4.5 also extends decision-tree learning by supporting numerical attributes and missing values and by incorporating pruning. Quinlan's C4.5 work explicitly addresses unknown attribute values and overfitting-related issues. (ScienceDirect)

Main characteristics

- Uses **Gain Ratio**
- Handles categorical and numerical attributes
- Handles missing values
- Uses pruning
- Produces interpretable decision trees
- Can also represent learned knowledge as rules

C4.5 remains an important reference point for modern decision-tree research. A 2025 Scientific Reports study specifically investigates parameter selection and optimization for C4.5, emphasizing that parameter adjustment and pruning remain relevant to improving performance and generalization. (Nature)

3.3 CART

Classification and Regression Trees (CART) was introduced by Breiman, Friedman, Olshen and Stone. CART differs from ID3 and C4.5 by constructing **binary trees**, meaning that each internal node produces two branches. (WorldCat)

For classification, CART commonly uses the **Gini Index** as its splitting criterion. Research comparing splitting criteria describes CART as a binary-tree method based on Gini impurity. (DOI)

Main characteristics

- Uses **Gini Index** for classification
- Produces binary trees
- Supports classification and regression
- Can handle numerical and categorical variables
- Uses pruning through a complexity-based approach
- Provides highly interpretable rules

CART's binary structure can make the resulting tree more systematic, although it can also require additional splits compared with a multiway tree.

4. Comparative Analysis

4.1 ID3 vs C4.5 vs CART

Parameter	ID3	C4.5	CART
Full Name	Iterative Dichotomiser 3	C4.5	Classification and Regression Trees
Main Developer	J. Ross Quinlan	J. Ross Quinlan	Breiman, Friedman, Olshen & Stone
Main Criterion	Information Gain	Gain Ratio	Gini Index
Impurity Concept	Entropy	Entropy + Split Information	Gini impurity
Tree Structure	Multiway	Multiway	Binary
Numerical Data	Limited/basic ID3	Yes	Yes
Missing Values	Limited	Yes	Supported through implementation strategies
Pruning	No basic pruning	Error-based pruning	Cost-complexity pruning
Classification	Yes	Yes	Yes
Regression	No	Mainly classification	Yes
Interpretability	Very high	Very high	Very high
Overfitting Control	Weak in basic ID3	Stronger	Strong
Attribute Bias	Higher	Reduced using Gain Ratio	Different bias characteristics
Computational Complexity	Relatively low	Higher than ID3	Efficient but depends on split search
Main Strength	Simple and fast	Flexible and robust	Versatile and statistically grounded
Main Weakness	Limited data handling	More complex	Binary splits can create larger structures

The basic differences in splitting criteria and data-handling capabilities are supported by comparative decision-tree research. (DOI)

4.2 Attribute Selection

ID3 — Information Gain

Information Gain measures the reduction in entropy produced by splitting on an attribute.

Advantage: Simple and effective.

Limitation: Can favor attributes with many distinct values. (DOI)

C4.5 — Gain Ratio

Gain Ratio modifies Information Gain using split information.

Advantage: Reduces the many-valued-attribute bias associated with Information Gain.

Limitation: The measure itself can have undesirable behavior in certain situations, including sensitivity to low split-information values. (DOI)

CART — Gini Index

CART chooses splits that reduce Gini impurity.

Advantage: Computationally convenient and effective for classification.

Limitation: The binary splitting structure can lead to deeper trees in some datasets.

5. Pruning Strategies

Pruning is essential because an unrestricted decision tree can memorize training data and perform poorly on unseen data.

ID3

Basic ID3 does not contain the sophisticated pruning strategy associated with C4.5 and CART. This makes uncontrolled ID3 trees more vulnerable to overfitting. (PubMed Central (PMC))

C4.5

C4.5 uses **error-based pruning** after tree growth. This removes branches that do not provide sufficient predictive value and helps improve generalization. (PubMed Central (PMC))

CART

CART uses a **complexity-based pruning strategy**, producing a sequence of smaller subtrees and selecting an appropriate subtree based on predictive error and complexity. (PubMed Central (PMC))

Recent research demonstrates that pruning remains an active research problem. A 2025 ICML paper found that some optimal pruning operations can be solved in polynomial time, whereas others are NP-complete, showing that finding the optimal simplified tree can itself be computationally difficult. (Proceedings of Machine Learning Research)

6. Critical Analysis

ID3

Strengths

- Simple algorithm
- Easy to understand
- Fast tree construction
- Highly interpretable
- Provides a clear introduction to decision-tree learning

Weaknesses

- Information Gain can favor high-cardinality attributes.
- Basic ID3 has limited handling of numerical attributes.
- Basic ID3 lacks advanced pruning.
- Performance can deteriorate with noisy or incomplete data.

A multi-dataset experimental study reported that ID3 builds trees quickly and can produce short trees, while also identifying limitations involving categorical data, missing values and attribute-selection bias. (DOI)

C4.5

Strengths

- Improved attribute selection through Gain Ratio
- Handles numerical attributes
- Handles missing data
- Includes pruning
- Highly interpretable
- More flexible than basic ID3

Weaknesses

- More computationally involved than ID3
- Gain Ratio is not universally optimal
- Tree construction and parameter selection can still require careful tuning
- Can still overfit without appropriate controls

Recent C4.5 research continues to focus on parameter optimization, confirming that the basic algorithm still has room for improvement. (Nature)

CART

Strengths

- Supports classification and regression
- Uses Gini-based splitting for classification
- Handles numerical features effectively
- Binary structure is systematic
- Strong pruning framework
- Widely used in practical machine-learning systems

Weaknesses

- Binary splitting can produce deeper trees
- Individual trees can be unstable
- Small changes in training data can sometimes produce different structures
- Gini-based selection is not necessarily optimal for every dataset

A detailed analysis of decision-tree splitting methods reports that CART handles both categorical and numerical values and supports classification and regression, while also noting instability and computational considerations. (DOI)

7. Evidence-Based Comparative Validation

A particularly useful experimental study by Aaboub, Chamlal and Ouaderhman compared ID3, C4.5 and CART along with three other decision-tree methods using **12 UCI datasets**. The study evaluated classification accuracy, tree depth, number of leaf nodes and tree-development time, using five-fold cross-validation with repeated runs. (DOI)

Reported average classification accuracy

Algorithm	Average Accuracy
ID3	**84.74%**
CART	84.01%
C4.5	82.93%
PCC-Tree	82.63%
DR	80.73%
FWDT	79.90%

The published results therefore do **not** support the idea that C4.5 is always the most accurate. In this particular experimental framework, ID3 obtained the highest average accuracy, closely followed by CART. (DOI)

For example, on the Breast Cancer W-D dataset, ID3 achieved 93.06%, C4.5 92.61%, and CART 92.66%. On the Pima dataset, ID3 achieved 74.80%, CART 74.48%, and C4.5 71.15%. However, on the Wine dataset C4.5 achieved 93.06%, compared with 90.28% for ID3 and 88.39% for CART. (DOI)

Interpretation

This evidence demonstrates an important point:

There is no universally best decision-tree algorithm.

Performance depends on the dataset, feature characteristics, class distribution, splitting criterion, tree structure and pruning configuration.

8. Recent Research Developments

Recent research shows that decision-tree research has moved beyond simply comparing ID3, C4.5 and CART.

2024 — Decision Tree Survey

Mienye and Jere provided a broad survey of decision-tree concepts, algorithms and applications, covering ID3, C4.5, CART and later tree-based ensemble methods. The study emphasizes the continuing importance of interpretability while examining modern applications such as medical diagnosis and fraud detection. (Directory of Open Access Journals)

2024 — Noisy Data

Research on classification performance under noisy class variables specifically examines C4.5 and CART because robustness to incorrect class labels is important for real-world datasets. (Wiley Online Library)

2024 — Fairness

A 2024 Applied Soft Computing study proposed **Fair-C4.5**, incorporating individual and group fairness considerations into tree construction. The study reports improvements in fairness without reducing predictive accuracy in its experiments. (ScienceDirect)

2024 — Semi-Supervised CART

Research has also investigated semi-supervised extensions of CART, demonstrating that decision-tree methods can be adapted to situations where only part of the available data is labeled. (DOI)

2025 — C4.5 Optimization

A 2025 Scientific Reports study investigated parameter selection and optimization for C4.5, showing that tuning remains an important research direction. (Nature)

2025 — Optimal Pruning

A 2025 ICML study investigated the computational complexity of optimal decision-tree pruning. Its findings demonstrate that some pruning problems are tractable while others can be computationally hard. (Proceedings of Machine Learning Research)

9. Research Gap

The literature reveals several unresolved issues.

Gap 1: Accuracy vs Interpretability

Many modern improvements focus on increasing predictive performance, but larger or more complicated trees can reduce human interpretability.

Gap 2: Static Splitting Criteria

Traditional algorithms generally use one primary criterion:

- ID3 → Information Gain
- C4.5 → Gain Ratio

- CART → Gini Index

The best criterion can vary according to the dataset. Experimental evidence demonstrates that different algorithms perform differently across datasets. (DOI)

Gap 3: Pruning Complexity

Finding an optimal small tree is not always computationally easy. Recent pruning research demonstrates that different pruning operations have substantially different computational complexity. (Proceedings of Machine Learning Research)

Gap 4: Robustness

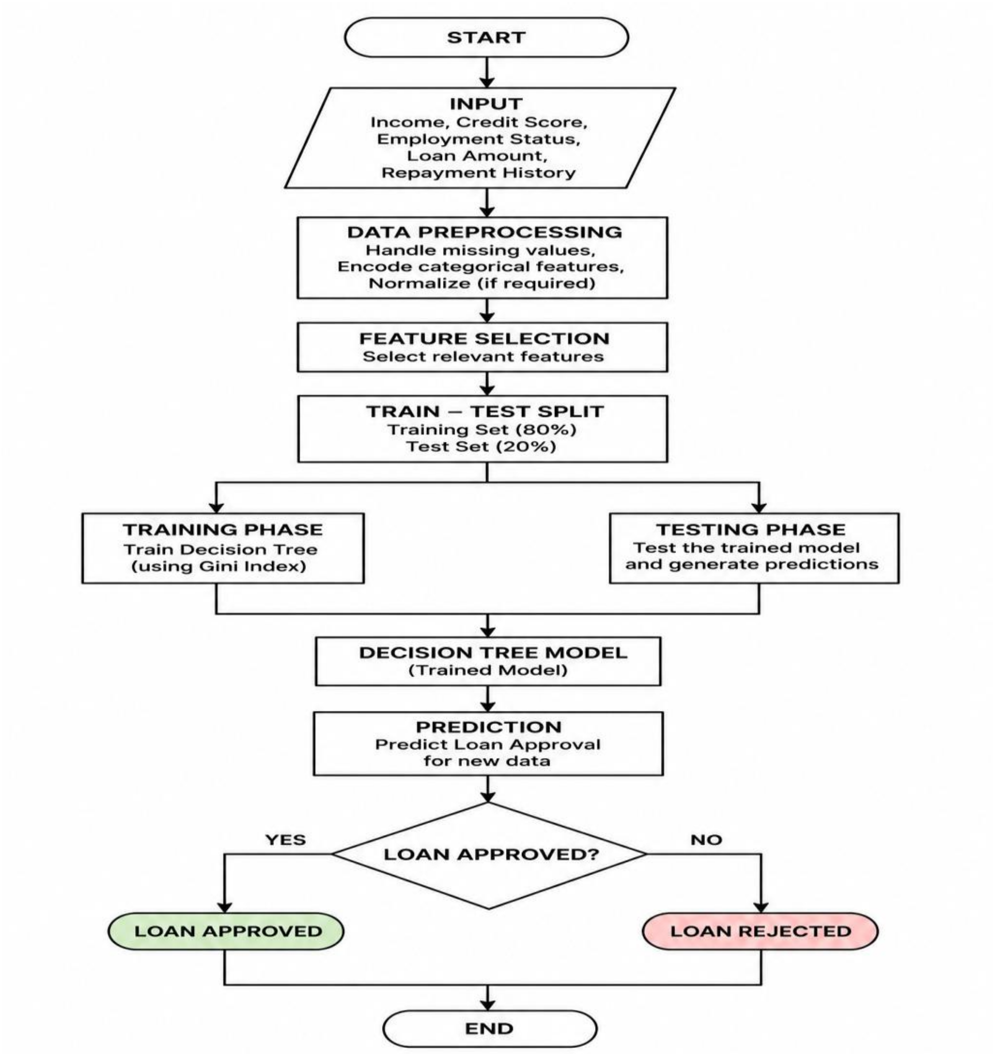
Real-world data may contain:

- Missing values
- Noise
- Incorrect labels
- Class imbalance
- Outliers**10. Proposed Research Direction**

Adaptive Multi-Objective Decision Tree

The proposed research direction is an **Adaptive Multi-Objective Decision Tree (AMODT)**.The system would dynamically select the most suitable splitting and pruning strategy based on the characteristics of the dataset.

Proposed Architecture



Proposed Objectives

The model would simultaneously optimize:

1. Classification accuracy
2. Tree depth
3. Number of leaf nodes
4. Training time
5. Robustness to noisy labels
6. Fairness across relevant groups
7. Interpretability

This direction builds on current research concerning splitting criteria, fairness, parameter optimization and pruning complexity. (DOI)

11. Research Methodology / Analysis Framework

The proposed research can follow this framework:

Research Question
↓
Literature Collection
↓
Dataset Selection
↓
Data Preprocessing
↓
Implement ID3
↓
Implement C4.5
↓
Implement CART
↓
Apply Proposed Adaptive Method
↓
Cross-Validation
↓
Performance Measurement
↓
Statistical Significance Testing
↓
Accuracy + Complexity + Fairness
↓
Comparative Analysis
↓
Research Conclusions

Evaluation Metrics

Category	Metric
Classification	Accuracy
Classification	Precision
Classification	Recall
Classification	F1-score
Complexity	Tree Depth
Complexity	Number of Leaves
Efficiency	Training Time
Robustness	Performance under Noise
Fairness	Group Fairness Measures
Interpretability	Tree Size / Rule Complexity

12. Experimental / Evidence-Based Validation

The proposed study would compare:

- ID3
- C4.5
- CART
- Adaptive Multi-Objective Decision Tree

using multiple datasets rather than relying on a single dataset.

Experimental Design

Experiment	Purpose
Experiment 1	Compare normal classification accuracy
Experiment 2	Compare tree depth
Experiment 3	Compare number of leaf nodes
Experiment 4	Compare training time
Experiment 5	Add noisy labels
Experiment 6	Evaluate class imbalance
Experiment 7	Evaluate fairness
Experiment 8	Compare pruning strategies

The existing 2023 study provides a strong basis for this experimental design because it already evaluated ID3, C4.5 and CART over multiple UCI datasets using accuracy, depth, leaf count and construction time. (DOI)

13. Result Analysis and Interpretation

The literature indicates that ID3, C4.5 and CART have different strengths rather than one universally dominant algorithm.

ID3 can be highly competitive in classification accuracy and tree construction speed, but its basic formulation has important limitations in handling numerical data, missing values and attribute-selection bias. (DOI)

C4.5 provides a more flexible development over ID3 through Gain Ratio, numerical attribute handling and pruning. However, its additional functionality introduces greater algorithmic complexity and does not guarantee the highest accuracy on every dataset. (PubMed Central (PMC))

CART provides a strong general-purpose framework because it supports both classification and regression and uses binary recursive partitioning. Its performance is competitive with ID3 and C4.5, but tree instability and binary splitting remain considerations. (DOI)

The experimental evidence is particularly important because the 12-dataset study found average accuracies of **84.74% for ID3, 84.01% for CART and 82.93% for C4.5**, demonstrating that algorithm selection should depend on the characteristics and objectives of the application rather than on assuming one algorithm is always superior. (DOI)

14. Discussion

The literature review demonstrates that decision trees remain valuable because they provide a combination of predictive capability and human-readable reasoning.

The historical progression can be summarized as:

```
ID3
│
│ Information Gain
▼
C4.5
│
│ Gain Ratio + Missing Values
│ + Numerical Attributes + Pruning
▼
CART
│
│ Gini + Binary Splitting
│ + Classification/Regression
▼
Modern Decision Trees
│
├── Fairness
├── Noise Robustness
├── Semi-Supervised Learning
├── Parameter Optimization
├── Advanced Pruning
└── Multi-Objective Optimization
```

15. Conclusion

This review examined **ID3, C4.5 and CART**, three foundational decision-tree learning approaches. ID3 introduced Information Gain and provided a simple and interpretable method for inductive classification. C4.5 improved upon ID3 through Gain Ratio, numerical-attribute handling, missing-value support and pruning. CART introduced binary recursive partitioning, Gini-based classification and a strong pruning framework. (DOI)

Comparative evidence demonstrates that no single algorithm consistently dominates across all datasets. A multi-dataset study found ID3 to have the highest average classification accuracy among the three, followed closely by CART, while C4.5 performed particularly well on some individual datasets. (DOI)

The major research gap is the lack of a unified approach that simultaneously considers **accuracy, robustness, fairness, computational efficiency and interpretability**. Recent research on fairness-aware C4.5, parameter optimization, semi-supervised CART and optimal pruning indicates that these are active and meaningful directions. (ScienceDirect)

Therefore, an **Adaptive Multi-Objective Decision Tree** is proposed as a future research direction. Such a model could dynamically select an appropriate splitting criterion and pruning strategy according to dataset characteristics while optimizing predictive performance and maintaining interpretable, robust and fair decision rules.

References

1. Quinlan, J. R. (1986). **Induction of Decision Trees.** *Machine Learning, 1*, 81–106. (DOI)
2. Quinlan, J. R. (1993). **C4.5: Programs for Machine Learning.** Morgan Kaufmann. (ScienceDirect)
3. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). **Classification and Regression Trees.** Wadsworth. (WorldCat)
4. Aaboub, F., Chamlal, H., & Ouaderhman, T. (2023). **Statistical analysis of various splitting criteria for decision trees.** *Journal of Algorithms & Computational Technology*. (DOI)
5. Mienye, I. D., & Jere, N. (2024). **A Survey of Decision Trees: Concepts, Algorithms, and Applications.** *IEEE Access, 12*, 86716–86727. (Directory of Open Access Journals)
6. **Classification Performance Analysis of Decision Tree-Based Algorithms with Noisy Class Variable.** *Discrete Dynamics in Nature and Society* (2024). (Wiley Online Library)

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.